∴ heat transferred from gas to surroundings is $W - \Delta U$

19 C

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left(\frac{d}{2}\right)^3 \Rightarrow V \propto d^3,\ V_f = \left(\frac{d_f}{d_i}\right)^3 V = \left(\frac{2d}{d}\right)^3 V = 8V$$

$$\Delta V = 8V - V = 7V$$

$$W = p\Delta V = 7pV$$

20 C